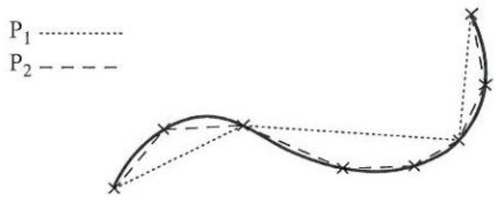
4.4: ARC LENGTH OF A CURVE

Introduction

1. One practical way to find the length of a curve between two of its points (for example to find the distance along the road between two towns, from a map) is to **mark several intermediate points along the arc**, **join them successively with straight lines** to form a polygonal line, and **measure the total length of this open polygon** to give an **approximation** to the length of the arc.

2. If you start with an open polygon P_1 and construct a new open polygon P_2 by **inserting additional points along the arc**, then P_2 will **fit better** than P_1, and the length of P_2 will be greater than the length of P_1 (see figure below).



3. In this way, you can form more and more open polygons with successively greater lengths. But since the shortest route between two points is the straight line joining them, the length of any such open polygon does not exceed the length of the curve.

   So you would expect that the lengths of successive approximations would be bounded above (by the length of the curve), and that **by putting the immediate points sufficiently close together** you could get **an approximation as close as you like to the length of the curve**.

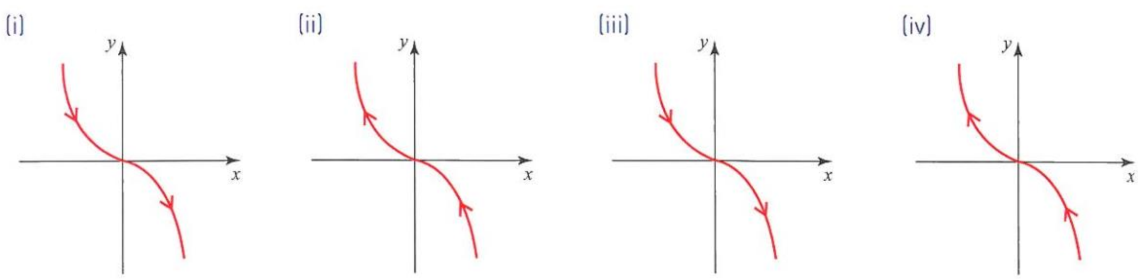
The Positive Sense Along a Curve [NIS]

1. If the coordinates of a point P on a curve are given in terms of a parameter p, then **the sense in which P moves along the curve as p increases** is called the **positive sense**.

2. The same applies if you are using x or y instead of p as the independent variable. The positive sense on a curve **depends on the particular way in which the Cartesian equation or parametric equations are expressed**.

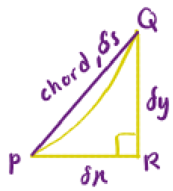
   For example, the equations
   (i) $y = -x^3$
   (ii) $x = -\sqrt[3]{y}$
   (iii) $x = p, y = -p^3$
   (iv) $x = -q, y = q^3$
   all give the same curve, using **independent variables**, x, y, q, p, respectively.

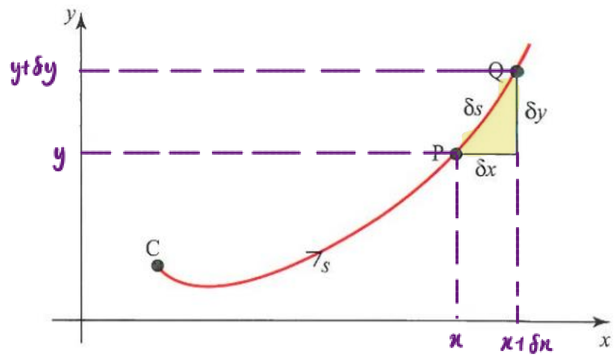
   In (i) and (iii), the positive sense is from left to right across the page.
   But in (ii) and (iv), the opposite sense is positive (see figure below).



Arc Length with Cartesian Coordinates

1. To see how to find the length of an arc, look at a part of a general curve, as shown in the figure below.





2. There, C is a fixed point on the curve, P is the point with parameter p, and s is the arc length from C to P, where $\boldsymbol{s}$ **is positive** if and only if **the motion from $\boldsymbol{C}$ to $\boldsymbol{P}$** is in the **positive sense along the curve**.

3. Let P and Q have coordinates (x, y) and $(x+\delta x, y+\delta y)$, corresponding to parametric values p and $p+\delta p$, respectively, and let (arc length PQ) = δs.

   The chord length PQ is $|PQ| = \sqrt{(\delta x)^2 + (\delta y)^2}$

   Just a fraction, not derivative

   Therefore $\frac{\delta s}{\delta p} = \frac{\delta s}{|PQ|} \times \frac{|PQ|}{\delta p}$

   $= \frac{\delta s}{|PQ|} \times \frac{\sqrt{(\delta x)^2 + (\delta y)^2}}{\delta p}$

   Note that $\delta p > 0$ [+ve sense]

   $\sqrt{(\delta p)^2} = |\delta p|$
   $= \delta p$

   $= \frac{\delta s}{|PQ|} \times \frac{\sqrt{(\delta x)^2 + (\delta y)^2}}{\sqrt{(\delta p)^2}}$

   $\boxed{\frac{\delta s}{\delta p} = \frac{\delta s}{|PQ|} \times \sqrt{\left(\frac{\delta x}{\delta p}\right)^2 + \left(\frac{\delta y}{\delta p}\right)^2}}$

   $\boxed{\lim_{\delta p \to 0} \frac{\delta s}{\delta p} = \frac{ds}{dp}}$

   $\therefore \frac{\delta x}{\delta p} \to \frac{dx}{dp}$

   $\frac{\delta y}{\delta p} \to \frac{dy}{\delta p}$

   $\delta s \to |PQ| \implies \frac{\delta s}{|PQ|} = 1$

   $\boxed{\frac{ds}{dp} = \sqrt{\left(\frac{dx}{dp}\right)^2 + \left(\frac{dy}{dp}\right)^2}}$

   $\downarrow$

   $\boxed{s = \int_{p_1}^{p_2} \sqrt{\left(\frac{dx}{dp}\right)^2 + \left(\frac{dy}{dp}\right)^2}\, dp}$

   As $\delta p \to 0$, $\frac{\delta s}{\delta p}, \frac{\delta x}{\delta p}, \frac{\delta y}{\delta p}$ tend to $\frac{ds}{dp}, \frac{dx}{dp}, \frac{dy}{dp}$, respectively. Also, $\frac{\delta s}{|PQ|} \to 1$.

   So taking limits as $\delta p \to 0$ gives the basic result:

$$\frac{ds}{dp} = \sqrt{\left(\frac{dx}{dp}\right)^2 + \left(\frac{dy}{dp}\right)^2}$$

   From this, $\boldsymbol{s}$ **can be found by integrating with respect to $\boldsymbol{p}$**.

4. If the independent variable is x (i.e. the equation of the curve is given in the form $y = f(x)$), then you put $p = x$ in the basic result. Then:

$$\frac{dx}{dp} = \frac{dx}{dx} = 1 \quad \text{and} \quad \frac{dy}{dp} = \frac{dy}{dx}$$

   so that:

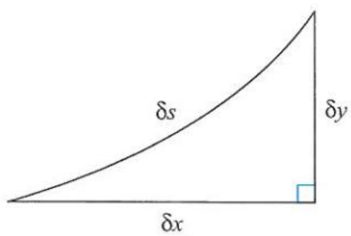
$$\frac{ds}{dx} = \sqrt{1 + \left(\frac{dy}{dx}\right)^2}$$

5. Similarly, when the independent variable is y,

$$\frac{ds}{dy} = \sqrt{\left(\frac{dx}{dy}\right)^2 + 1}$$

6. ***Memorisation Tips***:

All these results are easy to remember from the right-angled 'triangle' (see figure below) in which $(\delta s)^2 \approx (\delta x)^2 + (\delta y)^2$ by Pythagoras' theorem.



$$(ds)^2 = (dx)^2 + (dy)^2$$

$$\left(\frac{ds}{dp}\right)^2 = \left(\frac{dx}{dp}\right)^2 + \left(\frac{dy}{dp}\right)^2$$

The three results follow in the **limit** from **dividing by** $(\boldsymbol{\delta p})^2$**,** $(\boldsymbol{\delta x})^2$ **or** $(\boldsymbol{\delta y})^2$, as appropriate, and then taking the **positive square root** of each side. Notice that the positive root is needed in each case, since, by definition, s increases with the independent variable.

__Arc Length with Polar Coordinates__

1. The method of finding the length of a curve from its polar equation comes from differentiating the relations $x = r\cos\theta$, $y = r\sin\theta$ with respect to θ, remembering that r is a function of θ. Thus:

$$\frac{dx}{d\theta} = r(-\sin\theta) + \frac{dr}{d\theta}(\cos\theta) = -r\sin\theta + \left(\frac{dr}{d\theta}\right)(\cos\theta)$$

and

$$\frac{dy}{d\theta} = r\cos\theta + \left(\frac{dr}{d\theta}\right)(\sin\theta)$$

so that

$$\left(\frac{dx}{d\theta}\right)^2 = r^2\sin^2\theta + \left(\frac{dr}{d\theta}\right)^2\cos^2\theta - 2r\left(\frac{dr}{d\theta}\right)(\sin\theta)(\cos\theta) \quad \text{—— } ①$$

and

$$\left(\frac{dy}{d\theta}\right)^2 = r^2\cos^2\theta + \left(\frac{dr}{d\theta}\right)^2(\sin^2\theta) + 2r\left(\frac{dr}{d\theta}\right)(\sin\theta)(\cos\theta) \quad \text{—— } ②$$

Adding these and using $\cos^2\theta + \sin^2\theta = 1$ gives:

$$① + ② : \left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2 = r^2(\sin^2\theta + \cos^2\theta) + \left(\frac{dr}{d\theta}\right)^2(\cos^2\theta + \sin^2\theta)$$

$$\boxed{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2 = r^2 + \left(\frac{dr}{d\theta}\right)^2} \quad \text{—— } ⊛$$

$$\boxed{\left(\frac{ds}{d\theta}\right)^2 = \left(\frac{dr}{d\theta}\right)^2 + r^2}$$

so that

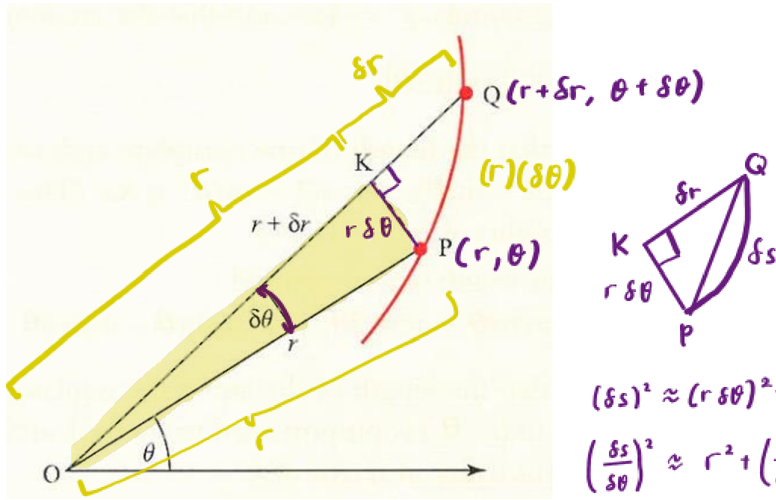
$$\boxed{\frac{ds}{d\theta} = \sqrt{\left(\frac{dr}{d\theta}\right)^2 + r^2},}$$

from which ***s* is found by integrating with respect to *θ***.

$$\boxed{s = \int_{\theta_1}^{\theta_2} \sqrt{\left(\frac{dr}{d\theta}\right)^2 + r^2}\, d\theta}$$

2. ***Memorisation Tips***:

The figure below shows a curve through neighbouring points P and Q with polar coordinates (r, θ) and $(r + \delta r, \theta + \delta\theta)$.

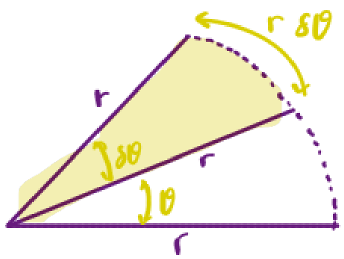


$(\delta s)^2 \approx (r\,\delta\theta)^2 + (\delta r)^2$

$\left(\frac{\delta s}{\delta\theta}\right)^2 \approx r^2 + \left(\frac{\delta r}{\delta\theta}\right)^2$

when $\delta\theta \to 0$

$\left(\frac{ds}{d\theta}\right)^2 = r^2 + \left(\frac{dr}{d\theta}\right)^2$



Example 10 (CIE 2018s11/1)

The curve C is defined parametrically by

$$x = e^t - t, \quad y = 4e^{\frac{1}{2}t}.$$

Find the length of the arc of C from the point where $t = 0$ to the point where $t = 3$. [5]

$(Ans: 2 + e^3)$

Solution

$x = e^t - t \quad , \quad y = 4e^{\frac{1}{2}t}$

$\frac{dx}{dt} = e^t - 1 \quad , \quad \frac{dy}{dt} = 2e^{\frac{1}{2}t}$

$\left(\frac{ds}{dt}\right)^2 = \left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2$

$= (e^t - 1)^2 + (2e^{\frac{1}{2}t})^2$

$= e^{2t} + 2e^t + 1$

$= (e^t + 1)^2$

$\frac{ds}{dt} = \sqrt{(e^t + 1)^2}$

$= e^t + 1$

$s = \int_0^3 (e^t + 1)\,dt$

$\vdots$

DIY